

(Use Lagrangian formalism to work out all the problems)

1. A particle of mass m is constrained to move on the inside surface of a smooth cone of half-angle α (see Fig. 1). The particle is subject to a gravitational force. Determine a set of generalized coordinates and determine the constraints. Find Lagrange's equations of motion.

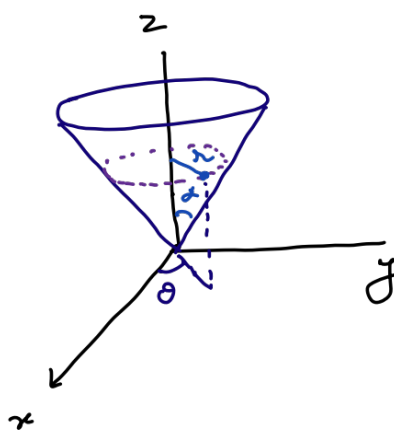


Fig. 1

2. Find the Lagrangian for the system shown in Fig. 2. The particle m_2 moves on a vertical axis and the whole system rotates about this axis with a constant angular velocity ω . Take θ as the angle between one of the segments a and the vertical, and ϕ the angle of rotation of the system about the axis. The system is in a earth's gravitational field.

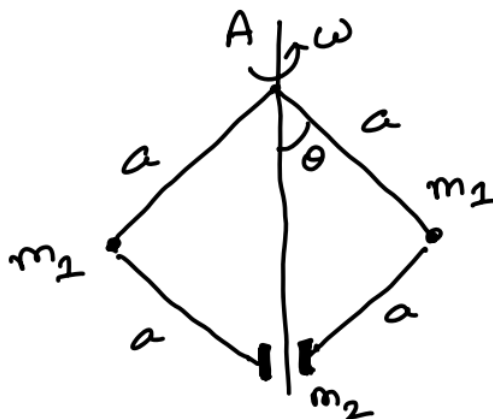


Fig. 2

3. A bead slides along a smooth wire bent in the shape of a parabola $z = cr^2$ (see Fig. 3). The bead rotates in a circle of radius R when the wire is rotating about its vertical symmetry axis with angular velocity ω . Find the value of c .

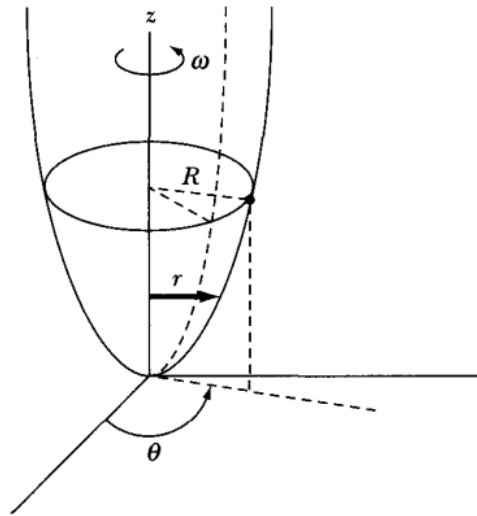


Fig. 3

4. Find the frequency of small oscillations of a simple pendulum placed in a railroad car that has a constant acceleration a in the x -direction. (Refer to Fig. 4)

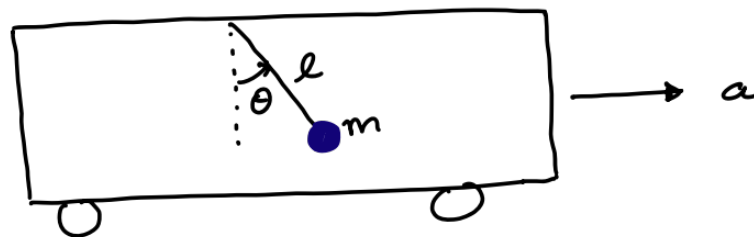


Fig. 4

5. A solid ball of mass m rolls down a movable wedge of mass M . The angle of the wedge is θ , and it is free to slide on a smooth horizontal surface. The contact between the ball and the wedge is perfectly rough. Find the acceleration of the wedge. (Refer to Fig. 5)

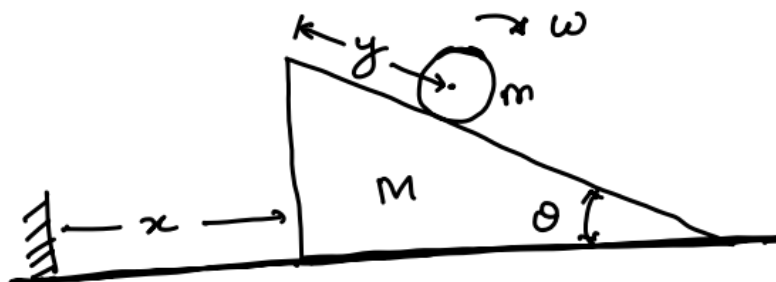


Fig. 5